

# WEEKLY PRACTICE AND TROUBLESHOOTING

**Topic:** WAN, VPN and NAT/PAT

## Question 1

Use Cisco Packet Tracer to simulate a WAN setup connecting two branch offices via a VPN tunnel and configure NAT on the edge routers so both branches can access the internet.

**Answer:** A WAN connects two branch offices. A VPN tunnel connects the branches securely, and NAT allows their private devices to access the Internet.

### WAN + VPN + NAT Practical Topology

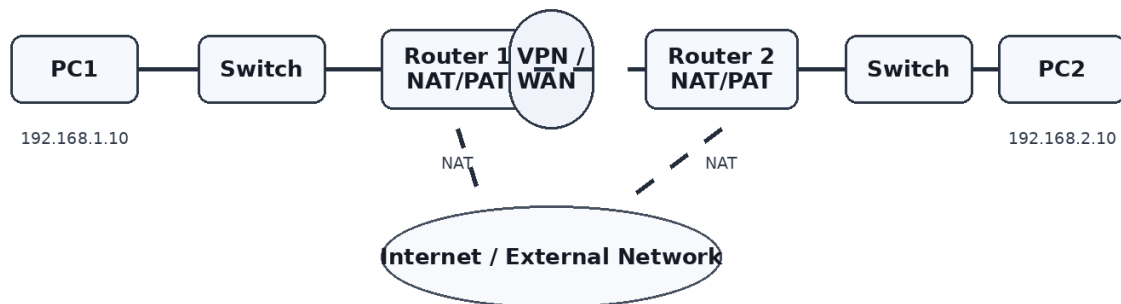


Figure 1: Basic WAN, VPN and NAT topology.

## Question 2

Configure PAT on a router so that three internal devices can browse the Internet using one public IP address.

**Answer:** PAT allows many private devices to share one public IP address.

```
access-list 1 permit 192.168.1.0 0.0.0.255
ip nat inside source list 1 interface g0/1 overload
```

## Question 3

Misconfigure the VPN pre-shared key and use **show crypto isakmp sa** and **show run** to find and fix the problem.

**Answer:** If the keys on the two routers are different, the VPN will not work. Check the configuration and set the same key on both routers.

```
show crypto isakmp sa
show run
crypto isakmp key cisco123 address 10.0.0.2
```

## Question 4

Troubleshoot a case where one branch can access the Internet but cannot access the other branch through the VPN.

**Answer:** Check the VPN status and routing table. If the route to the other branch is missing, add the correct route.

```
show crypto isakmp sa
show crypto ipsec sa
show ip route
```

```
ip route 192.168.2.0 255.255.255.0 10.0.0.2
```

Then test the connection with:

```
ping 192.168.2.10
```

## Question 5

Swap the WAN, VPN and NAT configuration with a classmate, find one mistake, and explain how you fixed it using show and debug commands.

**Answer:** I checked the configuration and found an incorrect NAT or routing setting. I corrected it and tested the connection again.

```
show run
show ip route
show ip nat translations
debug ip nat
```

**Conclusion:** WAN connects different networks, VPN connects branches securely, and NAT/PAT allows private devices to access the Internet.